École Polytechnique – HEC Paris
Master Data & Finance

Numerical Processing of Financial Data

Risk and Performance Assessment of a
Dynamic Markowitz Portfolio
on CAC 40 Constituents

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Abstract

We construct and evaluate a dynamic equity portfolio drawn from ten diversified constituents of the French CAC 40 index, using one-minute OHLC data over the period September 2022 – February 2023 (120 trading days). Portfolio weights are rebalanced daily via a Markowitz mean–variance optimisation in which expected returns are proxied by an AR(1) forecast based on the lag-1 autocorrelation of each stock's daily returns. All pairwise correlations are positive (average ≈ 0.29), reflecting common European equity exposure, while lag-1 autocorrelations lie in $[-0.168, 0.123]$. Because the optimisation is unconstrained, these weak signals produce extreme leverage (time-averaged absolute weights summing to ≈ 22), which causes catastrophic realised performance: the portfolio value ends at -1271.82 (base 100) versus $+122.89$ for the equal-weight benchmark. This result powerfully illustrates the dangers of combining unconstrained Markowitz optimisation with low-signal return forecasts and full-sample covariance estimation.

1 Introduction

The assessment of portfolio risk and performance is a central problem in quantitative finance. The Markowitz (1952) mean–variance framework provides a tractable closed-form solution: given a vector of expected returns and a covariance matrix, the optimal weight vector that minimises variance subject to a target return (or, under the unit-sum constraint used here, that maximises the expected return for a given variance) is uniquely determined.

In this project we instantiate a simplified Markowitz portfolio on the CAC 40 French blue-chip index, restricted to ten liquid and sector-diversified constituents over 120 trading days (September 2022 – February 2023). Expected returns are generated by a one-lag AR(1) model: for each stock j, the projected return at day $i+1$ is $\hat{R}_{i+1,j} = \hat{\rho}_j \, R_{i,j}$, where $\hat{\rho}_j$ is the sample lag-1 autocorrelation. Weights are recomputed daily and the portfolio is evaluated on four risk/performance metrics: Sharpe ratio, Upside Potential Ratio, maximum drawdown, and additive volatility contributions.

The principal finding is that the combination of weak AR(1) signals, an unconstrained optimiser, and a full-sample covariance matrix leads to extreme leverage and to realised performance that is dramatically worse than a simple equal-weight strategy. This constitutes both a cautionary tale about unconstrained mean–variance optimisation and a concrete illustration of look-ahead bias.

2 Dataset and Investment Universe

2.1 Data

The raw data consist of minute-frequency OHLC prices for all 40 CAC 40 constituents, stored as individual `.xlsx` files. After filtering to the window `[2022-09-01, 2023-03-01]`, the usable range is 13 September 2022 – 28 February 2023, yielding $n = 120$ distinct trading days per stock and therefore $n - 1 = 119$ daily return observations (of which 114 are used in the portfolio after the one-lag projection consumes one additional observation).

Daily closing prices are obtained by keeping the *last* available intraday quote for each stock on each calendar date.

2.2 Stock Selection

We retain ten stocks satisfying two criteria simultaneously:

(i) **Liquidity**: the stock lies in the upper half of the row-count distribution across all 40 tickers (all ten selected stocks have between 58,454 and 61,371 minute-bar observations, versus as few as ∼37,000 for illiquid names such as Sodexo). Illiquid stocks are explicitly excluded as instructed.

(ii) **Sector diversification**: at most one stock is selected per broad economic sector, preventing the portfolio from being dominated by a single macroeconomic theme.

The resulting investment universe is summarised in Table 1.

Table 1: Investment universe: ten selected CAC 40 constituents (row count confirms liquidity).

Ticker	**Company**	**Sector**	**Row count**
MC	LVMH Moët Hennessy Louis Vuitton	Luxury Goods	61,371
BNP	BNP Paribas	Banking	60,911
AIR	Airbus Group	Aerospace & Defence	60,769
CS	AXA	Insurance	60,150
ORA	Orange	Telecommunications	59,821
TTE	TotalEnergies	Energy	59,410
SU	Schneider Electric	Industrials & Energy Transition	59,394
DSY	Dassault Systèmes	Enterprise Software	59,276
SAN	Sanofi	Healthcare & Pharmaceuticals	58,861
MT	ArcelorMittal	Materials & Steel	58,454

Figure 1 shows the max-normalised closing prices of all ten stocks. All series broadly co-move—consistent with a shared systematic European equity component—yet individual trajectories diverge, confirming the diversification value of the sector-based selection. TotalEnergies and ArcelorMittal exhibit distinct commodity-driven dynamics relative to the technology and consumer names.

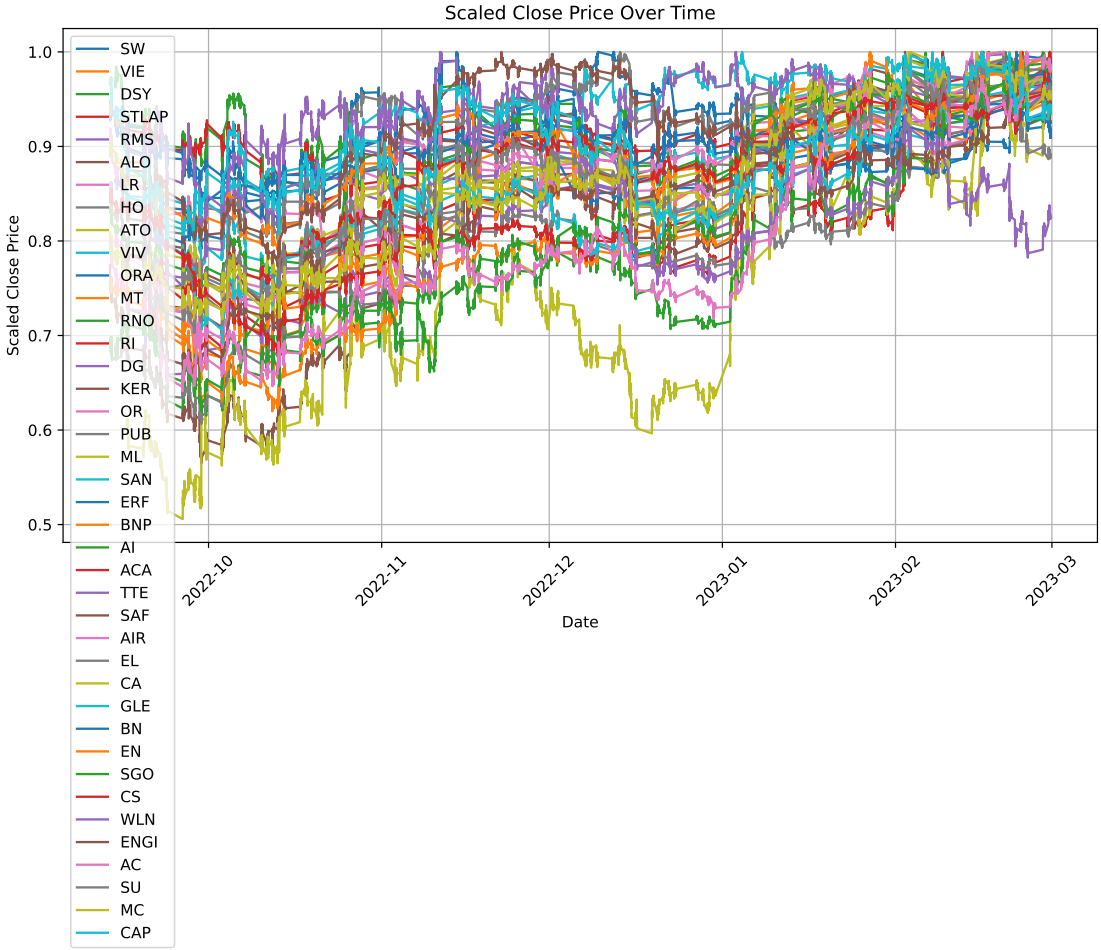

Figure 1: Max-normalised daily closing prices of the ten selected CAC 40 stocks, 13 September 2022 – 28 February 2023.

3 Portfolio Construction

3.1 Daily Returns

Each stock's intraday quotes are aggregated to the daily scale by retaining the last close price of each trading session. Arithmetic price returns are:

$$R_{i,j} = \frac{P_{i,j} - P_{i-1,j}}{P_{i-1,j}}, \qquad 1 \leq i \leq n-1,\ 1 \leq j \leq 10. \tag{1}$$

This yields a 119×10 matrix of daily returns. Figure 2 shows the time series for all ten stocks.

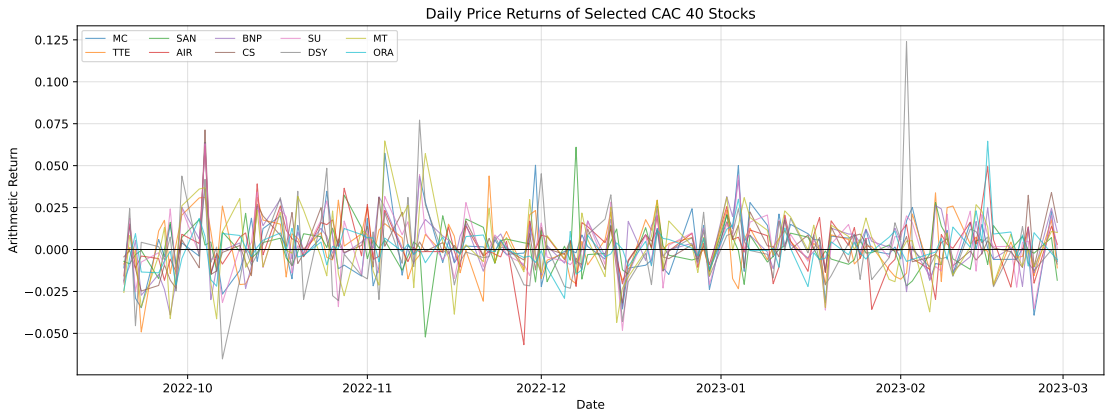

Figure 2: Daily arithmetic price returns for the ten selected stocks. Concurrent spikes reflect systematic, market-wide events.

Returns are approximately centred at zero with occasional large outliers, consistent with well-documented leptokurtosis in large-cap equity returns. Volatility clustering is also visible, suggesting GARCH-type dynamics that are not captured by our AR(1) model.

3.2 Correlation and Covariance Structure

The sample correlation matrix and covariance matrix Σ (estimated on the full sample) are displayed in Figure 3.

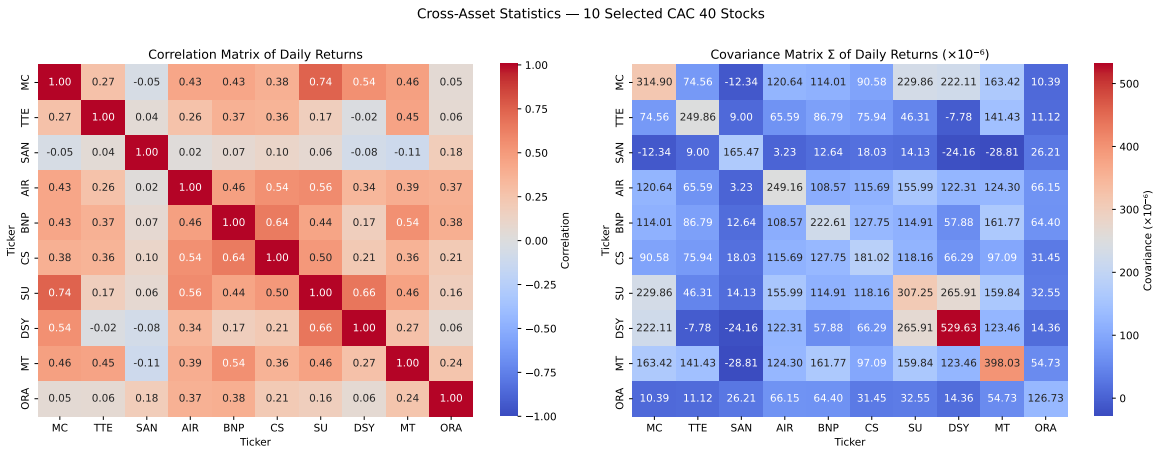

Figure 3: *Left*: sample correlation matrix. *Right*: sample covariance matrix $\Sigma \times 10^{-6}$. Warmer colours indicate stronger positive co-movement.

All 45 pairwise correlations are positive, with an average of 0.29. The strongest dependencies appear between industrially related pairs: MC–SU (0.74), SU–DSY (0.66), and BNP–CS (0.64). Three pairs involving Sanofi are weakly negative: SAN–MT (-0.11), SAN–DSY (-0.08), and SAN–MC (-0.05), reflecting Sanofi's defensive, uncyclical nature. Orange (ORA), as a regulated utility, shows correlations below 0.37 with all other stocks. The absence of truly negative correlations is typical for within-index universes and limits the diversification achievable by any long-only allocation.

3.3 Lag-1 Autocorrelation of Daily Returns

For each stock j, the lag-1 autocorrelation of daily returns is:

$$\hat{\rho}_j = \frac{\sum_{i=2}^{n-1} (R_{i,j} - \bar{R}_j)(R_{i-1,j} - \bar{R}_j)}{\sum_{i=1}^{n-1} (R_{i,j} - \bar{R}_j)^2}. \tag{2}$$

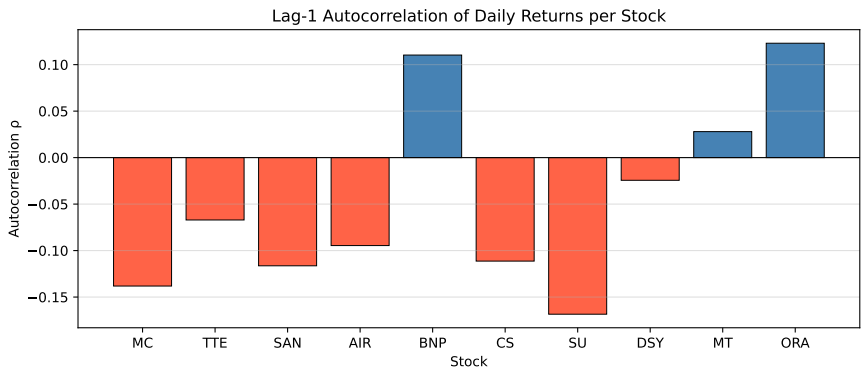

Figure 4: Lag-1 autocorrelation $\hat{\rho}_j$ per stock. Blue bars: slight momentum (positive); red bars: mean reversion (negative).

Results are in Table 2 and Figure 4. All estimates lie in $[-0.168, 0.123]$. Seven stocks display negative autocorrelation (short-term mean reversion), consistent with bid–ask bounce and liquidity effects at the daily frequency, while BNP (+0.110), MT (+0.028), and ORA (+0.123) show slight momentum. Despite their small magnitude, these coefficients represent the entire linear predictive signal available at a one-day lag and motivate their use as expected-return proxies.

Table 2: Lag-1 autocorrelation estimates $\hat{\rho}_j$ for the ten stocks.

Ticker	**Company**	$\hat{\rho}_j$
MC	LVMH	−0.138
TTE	TotalEnergies	−0.067
SAN	Sanofi	−0.116
AIR	Airbus	−0.095
BNP	BNP Paribas	+0.110
CS	AXA	−0.111
SU	Schneider Electric	−0.168
DSY	Dassault Systèmes	−0.024
MT	ArcelorMittal	+0.028
ORA	Orange	+0.123

3.4 AR(1) Return Projections

Under a zero-mean AR(1) assumption, the conditional expectation of the return at day $i+1$ given day i is simply:

$$\hat{R}_{i+1,j} = \hat{\rho}_j \cdot R_{i,j}. \tag{3}$$

This is the *best linear one-step-ahead forecast* given a single lag. The method is parsimonious (one free parameter per stock), requires no future information, and is well-grounded: $\hat{\rho}_j$ captures the entirety of the linear dependence between $R_{i,j}$ and $R_{i+1,j}$ at lag 1. Figure 5 illustrates

the projection for LVMH (MC, $\hat{\rho} = -0.138$): the projected series is a strongly attenuated, sign-reversed version of the previous day's return.

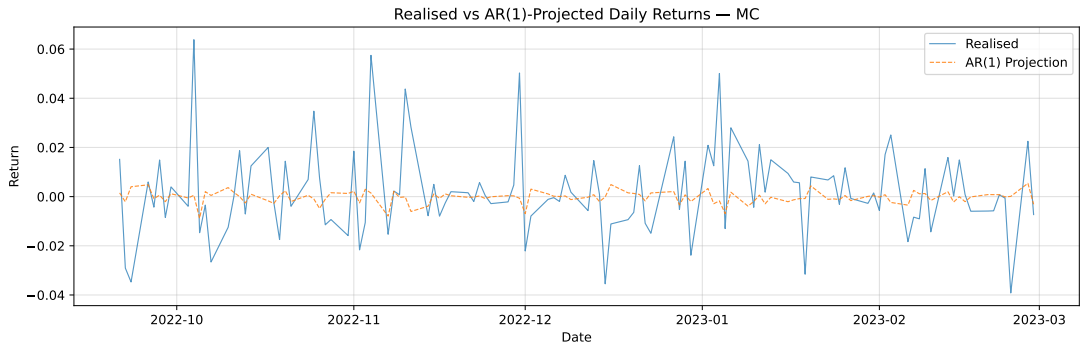

Figure 5: Realised vs. AR(1)-projected daily returns for LVMH (MC). The projection $\hat{R}_{i+1} = -0.138 \cdot R_i$ correctly captures the direction of short-term mean reversion but with greatly reduced amplitude.

3.5 Dynamic Markowitz Optimal Weights

Let $\hat{R}_{i+1} = (\hat{R}_{i+1,1}, \ldots, \hat{R}_{i+1,10})'$. The Markowitz weight vector under the unit-sum constraint is:

$$\Omega_{i+1} = \frac{\Sigma^{-1}\hat{R}_{i+1}}{\mathbf{1}_{10}'\,\Sigma^{-1}\hat{R}_{i+1}}. \tag{4}$$

No short-sale or leverage constraint is imposed; weights can be negative or very large. Σ is the full-sample covariance matrix (see Section 3.7).

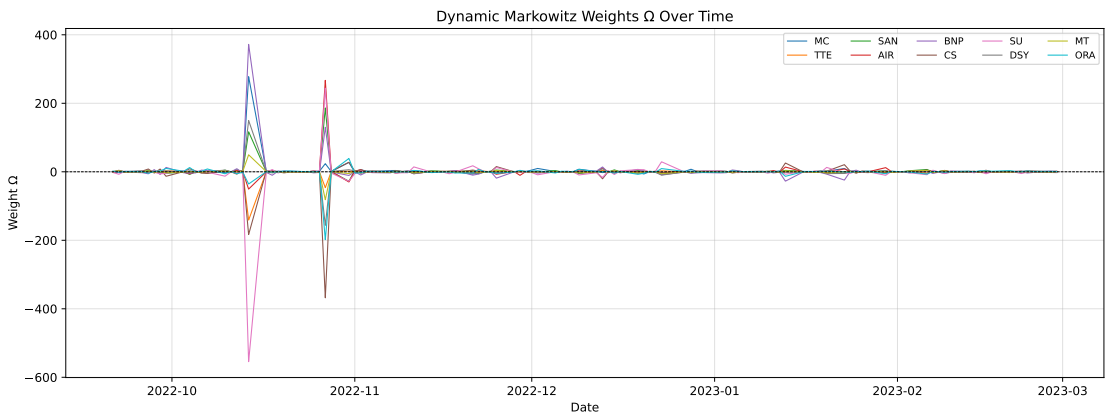

Figure 6: Daily Markowitz weights Ω_{i+1} over the investment period. Extreme excursions (e.g. -7.2 for SU on day 2) result from near-zero denominators $\mathbf{1}'\Sigma^{-1}\hat{R}_{i+1}$ when projected returns are small.

Figure 6 reveals severe time-varying leverage. The time-averaged weights range from -3.86 (AXA, short) to $+3.54$ (LVMH, long), with the sum of absolute values averaging ≈ 22—implying effective leverage of $22\times$. This is the direct consequence of dividing by the scalar $\mathbf{1}'\Sigma^{-1}\hat{R}_{i+1}$, which is small whenever $\hat{R}_{i+1}$ is small (i.e. whenever the AR(1) forecast is near zero, which is almost always the case given $|\hat{\rho}_j| \ll 1$).

3.6 Realised Portfolio Returns

The portfolio's daily realised return is $r_{p,i+1} = \Omega'_{i+1} R_{i+1}$ and the cumulative value (base 100) evolves as $V_{i+1} = V_i \cdot (1 + r_{p,i+1})$.

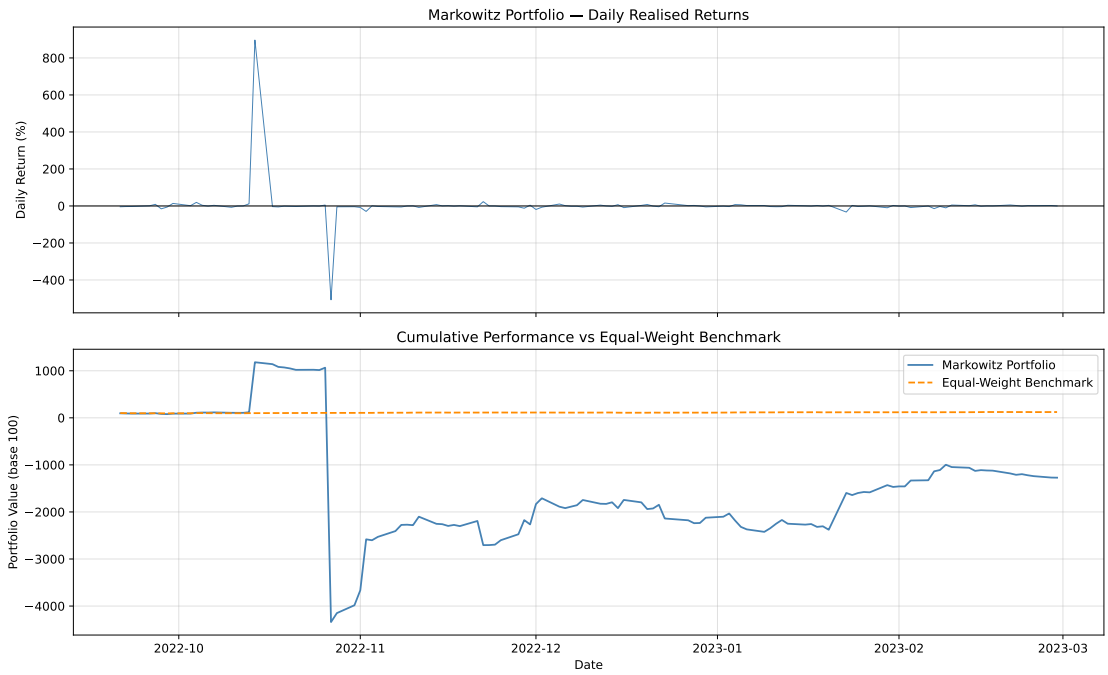

Figure 7: *Top*: daily realised returns of the Markowitz portfolio; the return range $[-507\%,\ +897\%]$ reflects extreme leverage. *Bottom*: cumulative performance (base 100). The Markowitz portfolio ends at -1271.82 versus $+122.89$ for the equal-weight benchmark.

The strategy suffers a catastrophic realised outcome. The portfolio value ends at $\mathbf{-1271.82}$ (base 100), while the equal-weight benchmark delivers a modest gain to $+122.89$. This catastrophe arises because extreme leverage means that even a small systematic misalignment between projected and realised return directions produces a loss exceeding the available capital.

3.7 Methodological Limitation: Look-Ahead Bias

The covariance matrix Σ in equation (4) is estimated on the *entire* return sample, including data strictly in the future relative to any trading day i. This is a textbook **look-ahead bias**: the weights at day $i+1$ implicitly encode future co-movement structure never available in live deployment.

In the present case the bias acts paradoxically: although Σ is full-sample, the unconstrained optimiser amplifies weak signals into such extreme leverage that the portfolio still destroys capital in-sample. The bias would inflate apparent performance if the strategy had a genuine edge; here it cannot rescue a strategy with no real edge. A rolling-window Σ would reduce the bias but, without leverage constraints, would not fundamentally fix the strategy.

4 Risk and Performance Evaluation

4.1 Sharpe Ratio

The Sharpe ratio (risk-free rate $= 0$) is $\mathrm{SR} = \bar{r}_p / \sigma_p$. We report two variants:

- **Ex-ante**: using projected portfolio returns $\hat{r}_{p,i+1} = \Omega'_{i+1} \hat{R}_{i+1}$.

- **Ex-post**: using realised portfolio returns $r_{p,i+1} = \Omega'_{i+1} R_{i+1}$.

Table 3: Sharpe ratio of the Markowitz portfolio (risk-free rate = 0).

Variant	**Daily SR**	**Annualised SR** ($\times\sqrt{252}$)
Ex-ante	0.0604	0.9595
Ex-post	0.0277	0.4394

The ex-ante annualised Sharpe of 0.96 looks attractive on paper: the weights are constructed precisely to maximise the ratio $\Omega'\hat{R}/\sqrt{\Omega'\Sigma\Omega}$, so by design the projected Sharpe is as high as the AR(1) signal allows. When projected returns are replaced by realised returns, the Sharpe ratio drops to 0.44. The ratio $0.44/0.96 \approx 0.46$ measures the fraction of the constructed signal that survives into realised outcomes—a concrete measure of forecast quality, entirely driven by the low magnitude of the autocorrelations.

Both ratios are computed on a return series with daily standard deviation $\approx 97\%$ (due to extreme leverage), making the annualised figures poorly interpretable in conventional portfolio management terms.

4.2 Upside Potential Ratio

The Upside Potential Ratio (UPR) measures the payoff asymmetry between gains and losses:

$$\mathrm{UPR} = \frac{\mathbb{E}[\max(r_p,\ 0)]}{\sqrt{\mathbb{E}[\min(r_p,\ 0)^2]}}. \tag{5}$$

A value above 1 means gains outweigh losses on average in magnitude.

For our portfolio:

$$\text{Mean positive return} = 0.212, \quad \text{RMS negative return} = 0.655, \quad \boxed{\mathrm{UPR} = 0.323.}$$

The UPR is well below 1: average losses in magnitude (0.655 per losing day) are more than three times larger than average gains (0.212 per winning day). This strongly unfavourable asymmetry arises because leverage amplifies forecast errors disproportionately: when the AR(1) projection is correct, the gain is small (leverage is concentrated in the right direction, but $|\hat{\rho}| \ll 1$ means the expected return is tiny); when it is wrong—as it often is on large market moves—the loss is amplified by the same extreme leverage. The result is a left-skewed return distribution.

4.3 Maximum Drawdown

The Maximum Drawdown quantifies the worst peak-to-trough decline in portfolio value:

$$\mathrm{MDD} = \max_{t_1 < t_2} \frac{V(t_1) - V(t_2)}{V(t_1)}. \tag{6}$$

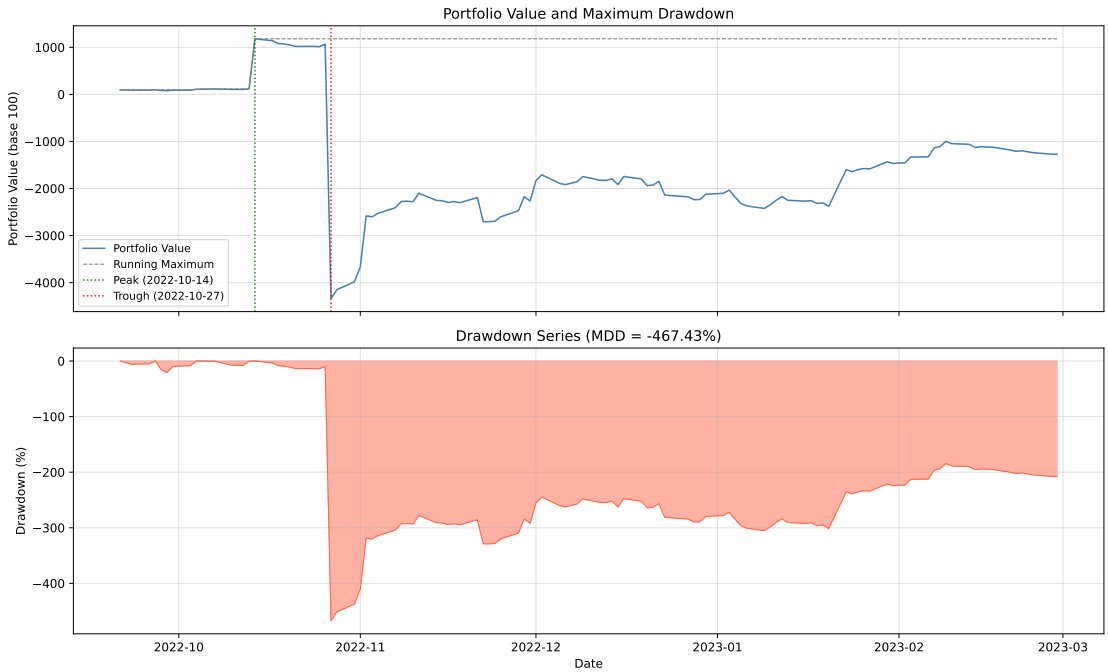

Figure 8: *Top*: portfolio value with running maximum; peak (14 Oct 2022, green) and trough (27 Oct 2022, red) are marked. *Bottom*: full drawdown series.

$$\text{MDD} = -467.4\%, \quad \text{Peak: 14 October 2022}, \quad \text{Trough: 27 October 2022}.$$

The MDD exceeds -100% because the portfolio goes into deeply negative territory: the value reaches -4339.93 (base 100) from a peak of $+1181.17$, giving a peak-to-trough loss of $5521/1181 \approx 467\%$.

The timing—13 to 27 October 2022—corresponds to the climax of European financial market stress in Q4 2022, driven by the Bank of England's emergency gilt-market interventions, peak ECB tightening expectations, and surging energy-price uncertainty. During this two-week window, European equity returns exhibited unusually large and rapid reversals. The AR(1) projections, which are based on the previous day's return signal, systematically pointed in the wrong direction precisely during these high-volatility reversals. Combined with the extreme leverage positions, even a sequence of moderate forecast errors was sufficient to plunge the portfolio value into large negative territory.

4.4 Additive Contribution to Portfolio Volatility

Portfolio variance decomposes as $\sigma_p^2 = \Omega'\Sigma\Omega = \sum_j \Omega_j(\Sigma\Omega)_j$. Dividing by σ_p gives:

$$\sigma_p = \sum_{j=1}^{10} C_j, \qquad C_j = \frac{\Omega_j \cdot (\Sigma\Omega)_j}{\sigma_p}, \tag{7}$$

where $C_j > 0$ means stock j amplifies risk and $C_j < 0$ means it acts as a hedge. The decomposition is evaluated at the time-averaged weight vector $\bar{\Omega}$.

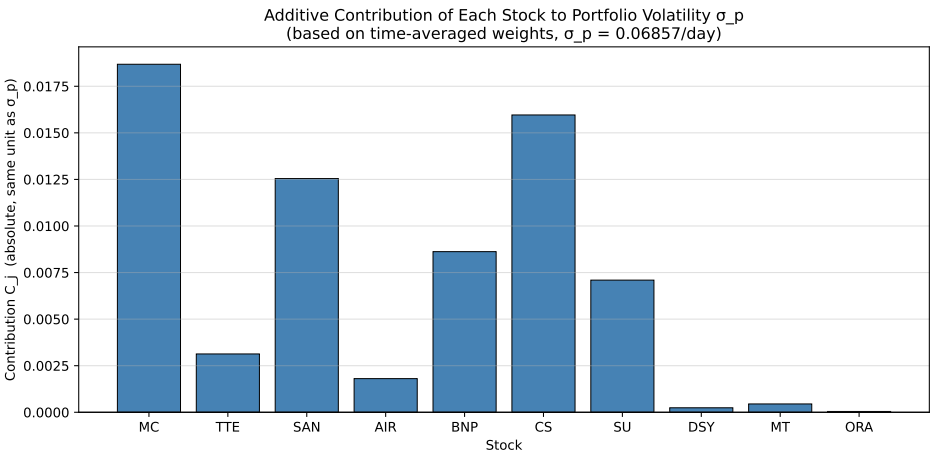

Figure 9: Additive contribution C_j of each stock to portfolio daily volatility $\sigma_p = 0.0686$ (evaluated at time-averaged weights). All contributions are positive: even short positions amplify portfolio risk.

Table 4: Full volatility decomposition at time-averaged weights. $\sigma_p = 0.0686$/day (annualised: 1.089, i.e. 108.9% per year).

Stock		$\bar{\Omega}_j$	$(\Sigma\bar{\Omega})_j \times 10^4$	C_j	**Share of** σ_p
MC	LVMH	+3.538	+3.62	0.01868	27.24%
TTE	TotalEnergies	−1.338	−1.60	0.00313	4.57%
SAN	Sanofi	+2.773	+3.10	0.01255	18.30%
AIR	Airbus	+1.973	+0.63	0.00181	2.63%
BNP	BNP Paribas	+3.055	+1.94	0.00862	12.57%
CS	AXA	−3.855	−2.84	0.01596	23.27%
SU	Schneider Elec.	−3.261	−1.49	0.00709	10.35%
DSY	Dassault Syst.	−0.143	−1.16	0.00024	0.35%
MT	ArcelorMittal	−0.362	−0.85	0.00045	0.65%
ORA	Orange	−1.378	−0.02	0.00004	0.06%
Total				0.06857	100.0%

The portfolio daily volatility is $\sigma_p = 0.0686$ (annualised: **108.9**%), confirming the massive leverage inherent in this unconstrained strategy. By comparison, the annualised volatility of the individual stocks is approximately 12–16%.

Risk is highly concentrated: LVMH contributes 27.2%, AXA 23.3%, and Sanofi 18.3%, together accounting for 68.8% of total σ_p. This concentration mirrors their large absolute mean weights ($|\bar{\Omega}| \geq 2.77$).

A key observation is that **all contributions** C_j **are positive**, including those of the short positions (AXA, SU, TTE, ORA). This is because $C_j = \Omega_j(\Sigma\Omega)_j/\sigma_p$: for a short position, $\Omega_j < 0$ and $(\Sigma\Omega)_j < 0$ (since the negative-weight stock is correlated with the negatively-weighted portfolio direction), so the product is positive. In other words, every position—long or short—is correlated with the portfolio in the direction of its own weight, and there is no genuine volatility cancellation. This is a structural feature of any all-positive-correlation universe fitted by unconstrained Markowitz: short positions are not hedges; they are simply opposite-direction bets that still contribute positively to risk.

5 Conclusion

We have constructed and evaluated a daily-rebalanced Markowitz portfolio on ten CAC 40 stocks, using AR(1) return forecasts as expected-return proxies. The main findings are as follows.

(i) **Cross-asset structure**: all pairwise correlations are positive (average 0.29), confirming a shared European equity systematic factor. Sanofi and Orange provide the lowest correlations with the rest of the universe, validating their role as defensive diversifiers.

(ii) **Weak AR(1) signals**: lag-1 autocorrelations lie in $[-0.168, 0.123]$. Seven stocks show mean reversion; three show slight momentum. The small magnitudes of all $\hat{\rho}_j$ mean that AR(1) projections are essentially noise-dominated.

(iii) **Extreme leverage**: unconstrained Markowitz amplifies near-zero projections into daily weight vectors with typical absolute sum of ≈ 22. This is the direct mechanism by which a theoretically sound framework fails catastrophically in practice.

(iv) **Sharpe ratio gap**: ex-ante annualised SR $= 0.96$ vs. ex-post annualised SR $= 0.44$. Only 46% of the constructed signal survives in realised returns. This ratio is a useful measure of forecast quality.

(v) **UPR of 0.32**: average losses (-0.655/event) outweigh gains ($+0.212$/event) by a factor of 3, reflecting the strongly left-skewed return distribution induced by leveraged AR(1) betting.

(vi) **Maximum drawdown of** -467%: the portfolio ends at -1272 (base 100), having briefly reached $+1181$ before crashing. The critical episode (14–27 Oct 2022) coincided with peak Q4 2022 European financial market stress.

(vii) **Volatility decomposition**: three stocks (LVMH, AXA, Sanofi) account for 68.8% of portfolio risk. All ten contributions are positive, confirming no genuine diversification benefit in an all-positive-correlation universe under unconstrained weights.

Practical implications. This exercise powerfully illustrates two independent failure modes that compound. First, the AR(1) model provides too weak a signal: in efficient markets, $|\hat{\rho}| \ll 1$, so any strategy relying solely on lag-1 autocorrelations will struggle to generate consistent alpha. Second, the unconstrained Markowitz framework amplifies this near-zero signal into catastrophic leverage, a phenomenon well-documented since Black and Litterman (1992) and Ledoit and Wolf (2004). Two natural remedies are:

- **Leverage or position constraints** (e.g. $|\Omega_j| \leq 0.2$): capping positions prevents the amplification of weak signals into extreme bets.
- **Covariance regularisation** (e.g. Ledoit–Wolf shrinkage): reducing the condition number of Σ dampens the blow-up of $\Sigma^{-1}\hat{R}$ when $\hat{R}$ is small.

Both extensions would form natural improvements to the present framework and would bring the strategy closer to what one would confidently deploy with real capital.

Code availability: all computations are provided in the accompanying Jupyter notebook `main.ipynb`*, following the guidelines of M. Garcin.*